

Telemetry and Anomaly Detection

Telemetry is the continuous stream of housekeeping data a CubeSat produces about its own health: voltages, currents, temperatures, wheel speeds, attitude error, and more. Every subsystem contributes parameters to this stream, which makes telemetry the single most useful signal operators have for understanding what is actually happening on orbit, and anomaly detection is the discipline of picking meaningful deviations out of that stream before they become functional failures.

What Telemetry Data Looks Like

A typical telemetry frame is a timestamped snapshot of dozens to hundreds of parameters, downlinked either continuously during a pass or accumulated onboard and dumped in bulk. Each parameter has an expected nominal range, and most flight software also flags values that cross a hard safety limit in real time, independent of any ground-side analysis.

Telemetry Parameters by Subsystem

Telemetry Parameter	Subsystem	Typical Nominal Range
Battery voltage	Power	6.4–8.4 V (2S Li-ion)
Battery temperature	Power / Thermal	0–35 °C
Reaction wheel speed	ADCS	0–6,000 rpm
Reaction wheel temperature	ADCS / Thermal	-10–45 °C
Attitude error	ADCS	< 1 degree (fine-pointing mode)
Downlink signal strength	Communications	Above link margin threshold

Statistical Anomaly Detection Basics

The simplest anomaly detection technique is thresholding: flagging any parameter that falls outside a predefined nominal range. Thresholding catches gross faults quickly but misses slower problems, since a value can drift steadily and still stay inside its safety limits for a long time. A moving average smooths out pass-to-pass noise and reveals that kind of gradual trend — comparing a parameter's recent moving average against its established baseline is often what catches a developing issue weeks before it would ever cross a hard threshold.

Figure: Sample telemetry time-series plot of reaction wheel temperature over several weeks, with an anomaly flagged where the moving average departs from baseline ahead of any threshold crossing.

Why It Matters Across Subsystems

Anomaly detection only works well when it looks across subsystems rather than at one parameter in isolation. A battery that runs warmer than its baseline might indicate a charging problem in the power subsystem, or it might simply reflect elevated thermal dissipation from a nearby reaction wheel running hot

due to bearing wear — the two look similar in isolation but call for very different responses. Because thermal dissipation from both the battery and the reaction wheels feeds into the same spacecraft thermal environment, correlating power, thermal, and ADCS telemetry together is often the only way to tell which subsystem is actually the source of a drift.

Communications telemetry plays a similar cross-checking role: a downlink whose signal strength is weaker than the link budget predicts for a given pass geometry can point to an antenna deployment issue, a pointing problem, or ground station misconfiguration, and ruling those out usually means comparing several passes' worth of link telemetry rather than judging a single contact.

Because battery behavior, reaction wheel health, thermal dissipation, and link budget performance all surface first as telemetry trends, this parameter set is what ties the power, thermal, ADCS, and communications subsystems together operationally — anomaly detection across that combined telemetry picture is usually the earliest warning a mission gets of a developing problem, well before it would otherwise be visible to any single subsystem team working in isolation.